1. Today, let's make an IP changer script that is more complicated and robust. We should check the user's permission before we start performing other tasks so that the user can continue to change the IP address. To check if the user has sudo permission, "id -u" command.

```
kmin@kali-rrc:~$ id
uid=1000(kmin) gid=1000(kmin) groups=1000(kmin),27(sudo)
kmin@kali-rrc:~$ sudo id
uid=0(root) gid=0(root) groups=0(root)
kmin@kali-rrc:~$ id -u
1000
kmin@kali-rrc:~$ sudo id -u
0
kmin@kali-rrc:~$
```

2. If you want to put the output of a command in the script you can write the command in $(command).

```
  GNU nano 5.2
#!/bin/bash

if [ $(id -u) -eq 0 ]
then
        echo "Running script as root. Continuing...."
else
        echo "Permission denied. Quitting...."
        exit
fi

echo "Rest of the script is running"
```

3. Here's what the output looks like if you run the command with or without sudo permission.

```
kmin@kali-rrc:~$ nano robust_ipchanger.sh
kmin@kali-rrc:~$ chmod +x robust_ipchanger.sh
kmin@kali-rrc:~$ ./robust_ipchanger.sh
Permission denied. Quitting....
kmin@kali-rrc:~$ sudo ./robust_ipchanger.sh
Running script as root. Continuing....
Rest of the script is running
kmin@kali-rrc:~$
```

4. If you want to see the list of possible interfaces, you can grep and cut the file to see.

```
kmin@kali-rrc:~$ grep static interfaces | cut -d " " -f 2
eth0
eth1
kmin@kali-rrc:~$
```

5. Now, let's put the command into our script.

```
#!/bin/bash

if [ $(id -u) -eq 0 ]
then
        echo "Script is now running as root. Continuing"
else
        echo "Permission error. Not running with sudo. Quitting."
        exit
fi

path=/home/jrice/interfaces

interfacelist=$(grep static $path | cut -d ' ' -f 2)
numinterfaces=$(grep static $path | cut -d ' ' -f 2 | wc -l)

echo "We found $numinterface possible static interfaces"
echo "Here are the available interfaces we found: $interfacelist"
```

6. Here's the result when you run the command.

```
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
Script is now running as root. Continuing
We found 2 possible static interface(s).
Here are the available interfaces we found: eth0
eth1
jrice@kali-rrc:~$
```

7. If you don't want "eth1" on another line, you can break the line break by echoing out the command.

```
echo "We found $numinterfaces possible static interface(s)."
echo "Here are the available interfaces we found: $(echo $interfacelist)"
```

8. Here's the result if you run the command.

```
jrice@kali-rrc:~$ nano robust_ipchanger.sh
jrice@kali-rrc:~$ sudo ./robust_ipchanger.sh
Script is now running as root. Continuing
We found 2 possible static interface(s).
Here are the available interfaces we found: eth0 eth1
```

9. Now, let's ask the user to enter one of the available interfaces.

```
read -p "Which static interface would you like to reconfigure: " interface
```

10. We need to make a loop to check what the user enters. To make this, we need a while loop to check every possible interface.

```
isvalid=false
counter=1

while [ $counter -le $numinterfaces ]
do
        if [ $interface == $(echo $interfacelist | cut -d ' ' -f $counter)
        then
                isvalid=true
        fi
        let counter=counter+1
done
```

11. To check if the user enter is valid or not, we also need to make an if statement outside of the loop.

```
if [ $isvalid == true ]
then
        echo "Valid interface selected: $interface"
else
        echo "Invalid interface selected: $interface Quitting."
        exit
fi
```

12. Here's the result of the command when you run the script.

```
kmin@kali-rrc:~$ nano robust_ipchanger.sh
kmin@kali-rrc:~$ sudo ./robust_ipchanger.sh
Running script as root. Continuing....
Here are the available interfaces found: eth0 eth1
We found 2 possible static interface(s)
Which interface would you like to reconfigure: eth0
Valid interface selected: eth0
kmin@kali-rrc:~$ sudo ./robust_ipchanger.sh
Running script as root. Continuing....
Here are the available interfaces found: eth0 eth1
We found 2 possible static interface(s)
Which interface would you like to reconfigure: eth1
Valid interface selected: eth1
kmin@kali-rrc:~$ sudo ./robust_ipchanger.sh
Running script as root. Continuing....
Here are the available interfaces found: eth0 eth1
We found 2 possible static interface(s)
Which interface would you like to reconfigure: hello
Invalid interface selected: hello. Quitting....
kmin@kali-rrc:~$
```

13. Now, we need to search for the part that we have to change. You can search for the line number of the “eth” you choose with the grep command and cut it based on “:”

```
jrice@kali-rrc:~$ grep -n eth0 interfaces
10:auto eth0
11:iface eth0 inet static
jrice@kali-rrc:~$ grep -n eth1 interfaces
15:iface eth1 inet static
jrice@kali-rrc:~$ grep -n eth1 interfaces | cut -d ':' -f 1
15
jrice@kali-rrc:~$ grep -n eth0 interfaces | cut -d ':' -f 1
10
11
```

14. Use the echo $(command) to remove the line break. Since we only want the last line if there is the first line or not, we can use the “rev” to reverse, cut based on the space, and then reverse it again. Now, you’ll have the last line number of whichever eth you choose even if it doesn’t have the auto line.

```
kmin@kali-rrc:~$ echo $(grep -n eth0 interfaces | cut -d ':' -f 1) | rev | cut -d ' ' -f 1 | rev
11
kmin@kali-rrc:~$ echo $(grep -n eth1 interfaces | cut -d ':' -f 1) | rev | cut -d ' ' -f 1 | rev
15
kmin@kali-rrc:~$
```

15. Let’s put it to our script and see the output.

```
line=$(grep -n $interface $path | cut -d ':' -f 1)
line=$(echo $line | rev | cut -d ' ' -f 1 | rev)

echo "The value of line is: $line"
```

```
kmin@kali-rrc:~$ nano robust_ipchanger.sh
kmin@kali-rrc:~$ sudo ./robust_ipchanger.sh
Running script as root. Continuing....
Here are the available interfaces found: eth0 eth1
We found 2 possible static interface(s)
Which interface would you like to reconfigure: eth0
Valid interface selected: eth0
The value of line is: 11
kmin@kali-rrc:~$ sudo ./robust_ipchanger.sh
Running script as root. Continuing....
Here are the available interfaces found: eth0 eth1
We found 2 possible static interface(s)
Which interface would you like to reconfigure: eth1
Valid interface selected: eth1
The value of line is: 15
kmin@kali-rrc:~$
```

16. Ask the user for the desire IP address and netmask. Replace the line using the sed command.

```
line=$(grep -n $interface $path | cut -d ':' -f 1)
line=$(echo $line | rev | cut -d ' ' -f 1 | rev)
let line=line+1

read -p "Please enter the desired IP address: " ip
read -p "Please enter the desired mask: " mask

sed -i "${line}s/.*/    address $ip/" $path
let line=line+1
sed -i "${line}s/.*/    netmask $mask/" $path
```

17. Let's write some final steps to restart the service, check if the restart worked and show the new IP address.

```
systemctl restart networking
if [ $? -eq 0 ]
then
        echo "Networking restart success!"
else
        echo "Problem....."
fi

ip addr show dev $interface
```

18. Here's what the output looks like when you change the actual interfaces file.

```
kmin@kali-rrc:~$ sudo ./robust_ipchanger.sh
Running script as root. Continuing....
Here are the available interfaces found: eth0
We found 1 possible static interface(s)
Which interface would you like to reconfigure: 192.168.1.137
Invalid interface selected: 192.168.1.137. Quitting....
kmin@kali-rrc:~$ sudo ./robust_ipchanger.sh
Running script as root. Continuing....
Here are the available interfaces found: eth0
We found 1 possible static interface(s)
Which interface would you like to reconfigure: eth0
Valid interface selected: eth0
The value of line is: 11
Please enter the desire IP address: 192.168.1.137
Please enterthe desire mask: 255.255.255.0
Network restart successful
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:00:30:f6:3a:9f brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.137/24 brd 192.168.1.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::200:30ff:fef6:3a9f/64 scope link tentative
       valid_lft forever preferred_lft forever
kmin@kali-rrc:~$
```